

Nandan Bharatkumar Parikh

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, GA

August 2025 – May 2027

- Relevant Coursework: Computer Vision, Machine Learning, Efficient ML

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science

Pilani, India

August 2019 – May 2023

- GPA: 8.8/10, Major GPA: 9.48/10
- Relevant Coursework: Deep Learning, Applied Statistical Methods, Image Processing, Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms, Compiler Construction

RESEARCH EXPERIENCE

Graduate Student Researcher

Georgia Institute of Technology (Bio-MIBLab, Advisor: Prof. May Wang)

August 2025 – Present

Atlanta, GA

- Investigating hierarchical editing mechanisms for concept bottleneck models to improve explainability in vision systems
- Developing methods to enable localized concept editing where only semantically related concepts are affected
- Conducting experiments on CUB and CelebA-HQ datasets for fine-grained visual recognition tasks

Undergraduate Research Assistant

Harvard University | Advisors: Prof. Donglai Wei, Richard Schalek

November 2022 – June 2023

Boston, MA

- Built end-to-end microscopy image deblurring pipeline with OpenCV, PyTorch, and CUDA to increase throughput
- Achieved 20x inference speedup over SOTA methods while maintaining quality (PSNR: 33.23 dB, SSIM: 0.8837)
- Work accepted for poster presentation at Microscopy and Microanalysis conference

Undergraduate Researcher – Underwater Image Enhancement

BITS Pilani (Supervisor: Prof. Pratik Narang)

January 2022 – May 2022

Pilani, India

- Developed unsupervised domain adaptation GAN architecture in PyTorch to translate RGB to hyperspectral images
- Introduced spectral profile optimization loss for improved translation quality
- Achieved PSNR of 17 dB, approaching state-of-the-art performance for underwater image enhancement

INDUSTRY EXPERIENCE

Software Engineer II (SDE-2)

Flipkart (Hotels Search)

July 2023 – July 2025

Bengaluru, India

- Architected and scaled Java/Spring Boot microservices to handle 5x traffic growth (5k → 24k RPS) via performance tuning, load testing, threadpool, and JVM optimization
- Designed RESTful APIs for hotel search, achieving 50% latency reduction (2s → 900ms p95) with multi-tier caching using Redis and Aerospike, reducing DB load by 70%
- Built asynchronous event-driven architecture with Kafka and GCP PubSub for real-time cache invalidation and data sync across services
- Implemented proactive cache warming with Quartz scheduler and MySQL-based job management, reducing price staleness by 18%
- Developed Elasticsearch search pipeline with custom ranking and query optimization, cutting zero-result searches by 35%
- Integrated LLM-powered natural language search via LangChain4j and vector databases on GCP with Docker/Kubernetes
- Promoted early from SDE-I for exceptional backend architecture and scalability contributions

Software Engineering Intern

Flipkart

June 2022 – August 2022

Bengaluru, India

- Built Python security automation framework integrating MobSF across internal mobile app pipeline
- Developed self-service web portal with Google Apps Script reducing manual security audits by 60%

TEACHING EXPERIENCE

Teaching Assistant – Neural Networks and Fuzzy Logic

August 2022 – December 2022

BITS Pilani

Pilani, India

- Curated and shortlisted research papers for student term projects in neural networks and fuzzy systems
- Mentored students in preparing research-based term papers and implementation projects

PROJECTS

LLM Quantization for Efficient Inference | *PyTorch, OPT-125M*

GitHub

- Implemented multi-bit post-training quantization with multiple schemes and granularities for model compression
- Achieved 4x memory reduction and 10% perplexity improvement using quantization-aware training over PTQ

Checkmate - Multiplayer Gaming Platform | *Node.js, Socket.io, Unity, C#*

GitHub

- Built real-time multiplayer platform with Node.js and Socket.io supporting concurrent PvP game sessions
- Implemented event-driven architecture for low-latency game state synchronization across 3 multiplayer games
- Developed Unity 3D environment with AI pathfinding and session management for seamless game transitions

Compiler for Custom Programming Language | *C*

GitHub

- Built full compiler in C with lexical analysis, parsing, AST generation, semantic analysis, and code generation
- Achieved 6th rank out of 80 teams in university competition through modular architecture design

Deep Q-Network for Pong | *PyTorch, OpenAI Gym*

GitHub

- Implemented DQN reinforcement learning agent to play Atari Pong using experience replay and target networks
- Trained agent to achieve human-level performance through convolutional neural network-based Q-learning

Water Quality Time Series Analysis | *Python, Pandas, Scikit-learn*

GitHub

- Built end-to-end pipeline for forecasting pollutants in water using time series decomposition and ADF stationarity testing
- Compared VARIMAX, Linear Regression, Random Forests, LSTMs, and Transformers, achieving best MAPE of 7.90%

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL, JavaScript, MATLAB

Backend & Architecture: Spring Boot, Microservices, RESTful APIs, Event-Driven Architecture, Multithreading

ML/AI Libraries: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Scikit-learn, LangChain4j

Data & Messaging: Kafka, Redis, Aerospike, Elasticsearch, MongoDB, MySQL, Qdrant, JDBC

Cloud & DevOps: GCP, Docker, Kubernetes, CI/CD, Git

PUBLICATIONS & PRESENTATIONS

Real-time Image Deblurring to Improve Throughput of Serial-Section Volume EM

2023

R. L. Schalek, N. Parikh, Y. Wu, J. W. Lichtman, and D. Wei

- Microscopy and Microanalysis, vol. 29, no. S1, pp. 988-989 (Poster Presentation)

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Association of Computing Machinery (ACM), BITS Pilani

Sophomore – Junior Year

Core Member, Game Development Lead

- Developed program connecting students with experts in niche technical fields, bridging gaps in research
- Organized alumni-led talks on diverse career paths, providing students with mentorship and insights
- Led creation of game showcased at university's technical fest APOGEE, enhancing visibility and engagement
- Recruited and mentored students in game development, building a sustainable technical community

AWARDS & HONORS

Qualified for Indian National Mathematical Olympiad (INMO), 2018

Ranked 6th out of 80 teams in Compiler Construction Competition, BITS Pilani, 2022

Promoted early from SDE-I to SDE-II at Flipkart for exceptional backend contributions, 2024

INVITED TALKS & PRESENTATIONS

"Vector Search and Semantic Retrieval Systems" – Flipkart Internal Tech Talk, 2025

"Contextual Ranking and Federation of Hotels at Scale" – Flipkart Internal Tech Talk, 2024